

AL58 - RESEARCH METHODOLOGY AND INTELLECTUAL PROPERTY RIGHTS

UNIT - II RESEARCH DESIGN CASE STUDY WORKSHEET

SECTION 1: SYSTEM VARIABLES

Identify and classify all system components and parameters for your experiment.

- **Independent Variable(s) (IV):** (What system settings or features are you directly manipulating?)
- **Dependent Variable(s) (DV):** (What performance or business metric are you measuring?)
- **Extraneous Variables:** (What external factors could introduce noise or unwanted variation?)
- **Confounded Variables:** (How will you prevent an extraneous factor from mixing with your IV?)

SECTION 2: RESEARCH HYPOTHESES

Formulate statistical assertions for your study.

- Null Hypothesis (H₀):
- Alternative Hypothesis (H₁):
- Directional Hypothesis:
- Non-Directional Hypothesis:

SECTION 3: GROUPS, TREATMENTS & CORE PRINCIPLES

Define your test setups and control mechanisms.

- Control Group: (What serves as your baseline system setup?)
- Experimental Group: (What new configuration is receiving the treatment?)
- Control Principle Application: (How do you keep all non-test conditions identical?)
- Randomization Strategy: (How are requests/users randomly allocated?)
- Replication Strategy: (How many test runs or user sessions will be sampled?)

SECTION 4: EXPERIMENTAL DESIGN IMPLEMENTATION

Select the experimental structure applicable to your case study.

- A. Single-Factor Design (CRD)
- B. Single Blocking Factor Design (RBD)
- C. Two-Blocking Factor Design (Latin Square Design)
- D. Multi-Factor Design (Factorial Design)

SECTION 5: CONCLUSION & DECISION

- Primary Result / Interaction Observed:
- Hypothesis Verdict:
- Final System / Design Recommendation: